Department of Information Technology

Faculty of Social Sciences

University of Kelaniya - Sri Lanka

INFO 14012 - Basic Mathematics & Statistics

Assignment 01

Objective:

The purpose of this assignment is to introduce students to basic data analysis techniques using SPSS. Students will independently select and download a dataset from an online source, such as Kaggle or any other dataset repository, analyze it, and interpret the results.

Dataset:

Students must download a dataset of their choice from an online source (e.g., Kaggle, UCI Machine Learning Repository, Data.gov).

The dataset should contain at least the following variables:

- A unique identifier (e.g., Customer ID)
- A numerical variable (e.g., Age, Purchase Frequency)
- A categorical variable (e.g., Gender)
- A satisfaction-related score (on a scale of 1-5)

Tasks:

Report Format:

1. **Introduction:** Provide a brief overview of the dataset, including its source, purpose, and key attributes. Describe the background of the dataset, Mention the industry or domain it pertains to (e.g., customer satisfaction in retail, employee performance in HR, patient records in healthcare).

2. **Objective:** Clearly define the specific goal of the analysis using SPSS. Identify the key aspects to be analyzed, such as identifying factors influencing customer satisfaction, detecting trends in purchasing behavior, or examining correlations between demographic factors and ratings. Specify the statistical methods or techniques to be used (e.g., descriptive statistics, correlation analysis, regression modeling) to achieve the objective.

3. **Methodology:** This section outlines the step-by-step process followed for data analysis, from data preparation to inferential analysis. Each step ensures the dataset is properly processed, visualized, and analyzed to derive meaningful insights.

   Data Preparation

   - Load the Dataset into SPSS – Import the selected dataset into SPSS for analysis. Ensure the data types (numerical or categorical) are correctly assigned.
   - Identify and Handle Missing Values – Use SPSS tools to check for missing values and decide on appropriate handling techniques, such as imputation or removal.
   - Check for Duplicate Values –Detect any duplicate records. Decide whether to keep, merge, or remove them based on the context and purpose of the analysis.
   - Check for Outliers – Utilize descriptive statistics (e.g., box plots) to detect and manage outliers that may skew the analysis.

   Descriptive Statistics

   - Numerical Variables:
     - Compute measures of central tendency (mean, median, mode) and dispersion (range, variance, standard deviation) to understand data distribution.
   - Categorical Variables:
     - Create frequency tables to examine the distribution of categories within the dataset.

Data Visualization

- Bar Chart: Represent categorical data to compare frequencies of different groups.
- Histogram: Display the distribution of numerical variables.
- Pie Chart: Illustrate proportions of categorical variables for better visualization of percentage distributions.

Inferential Analysis

- Independent t-test: Conduct an independent t-test to determine if there is a statistically significant difference between two groups (e.g., male vs. female satisfaction scores).
- Correlation Analysis: Perform a correlation analysis (e.g., Pearson's correlation) to examine the relationship between numerical variables (e.g., age and satisfaction score).

Each step ensures a structured and systematic approach to analyzing the dataset, leading t to accurate and meaningful conclusions.

4. **Experiment and Results:** Present findings using tables and graphs.

5. **Conclusion:** Interpret the results in relation to the initial objective. Identify the key findings, explaining how the independent variables (e.g., age, gender, income level) influence the dependent variable (e.g., satisfaction score, purchase behavior). Summarize the main takeaways, highlight any limitations of the analysis, and suggest possible improvements or future research directions.

6. **References:** List all sources used, including dataset sources, academic papers, websites, or statistical guides. Follow a proper referencing style (e.g., APA, IEEE, or any format specified by your instructor). (if applicable).

Evaluation Criteria:

- **Data Preparation (10%)** – Accuracy in handling missing values and outliers.
- **Descriptive Statistics (20%)** – Correct computation and presentation.
- **Data Visualization (20%)** – Clarity and appropriateness of charts.
- **Inferential Analysis (20%)** – Proper selection and execution of tests.
- **Report Quality (30%)** – Coherence, structure, and interpretation.

Submission Guidelines:

- Submit a PDF report with your findings.
- Attach the SPSS output file (.sav and. spv) in a zip folder.
- Upload to the designated platform before the deadline.

…End of Assignment…